

Linear Algebra, MATH-4100 02, Fall 2025

Instructor:

Dan Stevenson (Dr. Stevenson), RCSID: steved8

Office Hours: See LMS

Grader:

Zach Wolski, RCSID: wolskz@rpi.edu

Textbooks:

- *Linear Algebra Done Right*, third edition by Sheldon Axler
- *Linear Algebra*, fourth or fifth edition by Stephen Friedberg et al
- *Introduction to Linear Algebra*, fifth edition by Gilbert Strang

While neither text is required for the course, practice problems may occasionally be drawn from either text.

Course Web Page: Check LMS

Student Learning Outcomes: Upon successfully completing this course, students will be able to demonstrate that they can

- Perform symbol manipulation on vectors and matrices
- Apply linear algebra techniques to solve problems arising in the applied sciences
- Construct formal proofs using deductive reasoning to solve theory-based problems in linear algebra
- Understand the relationship between vectors, matrices, linear transformations

Lectures: Before lecture, a document will be posted to LMS containing a list of topics to be covered and some suggested reading, as well as a list of in class exercises. Please review this document before attending lecture. At some point after lecture, a marked up version of the notes will be posted to LMS.

Problem sets: There will be 8-10 homework assignments available through Gradescope. They will be released each non-exam week on a Tuesday or Friday and due the following Tuesday or Friday by 11:59 pm EDT. The problem sets will consist of a combination of computational and theoretical problems. Some problems will require the student to construct a formal mathematical proof using deductive reasoning. Use of LaTeX or Overleaf is heavily encouraged! Please see LMS for more details.

The rules for collaborating on homework problems with others are as follows:

Before working with anyone, each student must read the **entire** homework problem set and (minimally) do the following

- Look up and memorize all relevant definitions and Theorems.
- Make an outline of what needs to be shown.
- Make an attempt at solving the problem.

Once students have satisfied the above criteria, students may seek help and collaborate with others (inside and outside of office hours) on figuring out how to write, refine and improve proofs using the following guidelines:

- Students may get help on problems from an instructor, TA or another student currently taking the course.
- Students may not get help on problems from anyone outside of the class without explicit permission from the instructors.
- Students are responsible to independently write their own submission from their own notes taken during any above interactions.

- Submissions with glaring similarities will violate academic integrity policies. The penalties for violating academic integrity policies are shown in a subsequent section on this syllabus.
- Students are responsible for protecting their work and preventing it from being copied by others. In the event of an academic integrity policy violation, all students involved will be held accountable and penalized.

Late Penalties Late homework will receive substantial penalties and will not be accepted after 3 days late. That being said, if you feel you may benefit from a short extension, you may email the instructor at least 48 hours BEFORE the due date to make such a request. Please detail while you believe an extension is warranted.

One day late (after 3am on Wednesday)	deduction of 10% of total points
Two days late (from 3am Thursday until 3am on Friday)	deduction of 20% of total points
Three days late (from 3am until the end of the day on Friday)	deduction of 30% of total points
Four days late	deduction of 40% of total points
More than three days late	no credit

Exams: The tentative dates of the exams will be

- Exam 1: 9/25
- Exam 2: 10/30
- Exam 3: 12/8

Final Exam: The final exam will be optional. Please see the grading scheme below

Grades: The items above will be weighted as follows:

Item	Scheme 1	Scheme 2
10 Homework Assignments	40%	28%
3 Exams	60% (20% each)	42% (14% each)
Optional Final Exam		30%

The final course score will be **maximum** of Schemes 1 and 2. Letter grades will be assigned based on the first applicable category below:

93% + is an A	90% + is an A-	
87% + is a B+	83% + is a B	80% + is a B-
77% + is a C+	73% + is a C	70% + is a C-
65% + is a D+	55% + is a D	
0% + is a F.		

Appeals: All appeals must be made within one week of the time the exam, quiz or homework grade is made available in Gradescope. You can appeal your grade either through Gradescope or by emailing Dr. Stevenson.

Absences: If you need to be absent for an exam or you will miss significant class time due to illness, school-sponsored travel or the like, please contact Dr. Stevenson as soon as possible.

Academic Integrity: Student-teacher relationships are built on trust. Acts which violate this trust undermine the educational process. The Rensselaer Handbook of Student Rights and Responsibilities defines various forms of Academic Dishonesty and you should make yourself familiar with these. Exams must be done individually and cheating will result in a grade of zero and a report to the Dean of Students. Students may work together on homework provided they follow the rules in the preceding section on homework. Students must work individually, without assistance on exam and quizzes. If rules are violated, students will receive a zero on the assignment, exam or quiz and a report will be made to the Dean of Students. Repeat offenders will fail the class.

DSS Statement: Rensselaer Polytechnic Institute strives to make all learning experiences as accessible as possible. If you anticipate or experience academic barriers based on a disability, please let me know immediately so that we can discuss your options. To establish reasonable accommodations, please register with The Office of Disability Services for Students. After registration, make arrangements with me as soon as possible to discuss your accommodations so that they may be implemented in a timely fashion. DSS contact information: dss@rpi.edu; 518-276-8197; 4226 Academy Hall.

Check LMS frequently throughout the semester for information and updates.